EDS 223 Homework 2: Environmental Justice Patterns

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Setup

```
# Loading packages
library(tidyverse)
library(sf)
library(tmap)
library(spData)
library(janitor)
library(here)
library(knitr)
```

```
tmap_mode("plot")
```

Part 1

```
#| output: false

# Reading in and cleaning data
# Reading and cleaning EJScreen data
ejscreen <- st_read(
  here("data",
    "ejscreen",
    "EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb")) |>
  clean_names()
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
  `/home/katefrankland/MEDS/eds223/EDS223-HW2/data/ejscreen/EJSCREEN_2023_BG_StatePct_with_AS
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS: WGS 84 / Pseudo-Mercator
```

```
# Reading and cleaning HOLC data
holc <- st_read(
  here("data",
    "mapping-inequality",
    "mapping-inequality-los-angeles.json")) |>
  clean_names()
```

```
Reading layer `mapping-inequality-los-angeles' from data source
  `/home/katefrankland/MEDS/eds223/EDS223-HW2/data/mapping-inequality/mapping-
inequality-los-angeles.json'
  using driver `GeoJSON'
Simple feature collection with 417 features and 14 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -118.6104 ymin: 33.70563 xmax: -117.7028 ymax: 34.30388
Geodetic CRS:  WGS 84
```

```
# Read in crashes data
crashes <- read.csv('data/SWITRS_LA/Crashes.csv')
```

```
# Organizing the grade values
holc <- holc |>
  mutate( grade = str_extract(as.character(grade), "^[ABCD]"),
    grade = factor(grade, levels = c("A", "B", "C", "D"))
  )
```

```
# Creating the HOLC Map

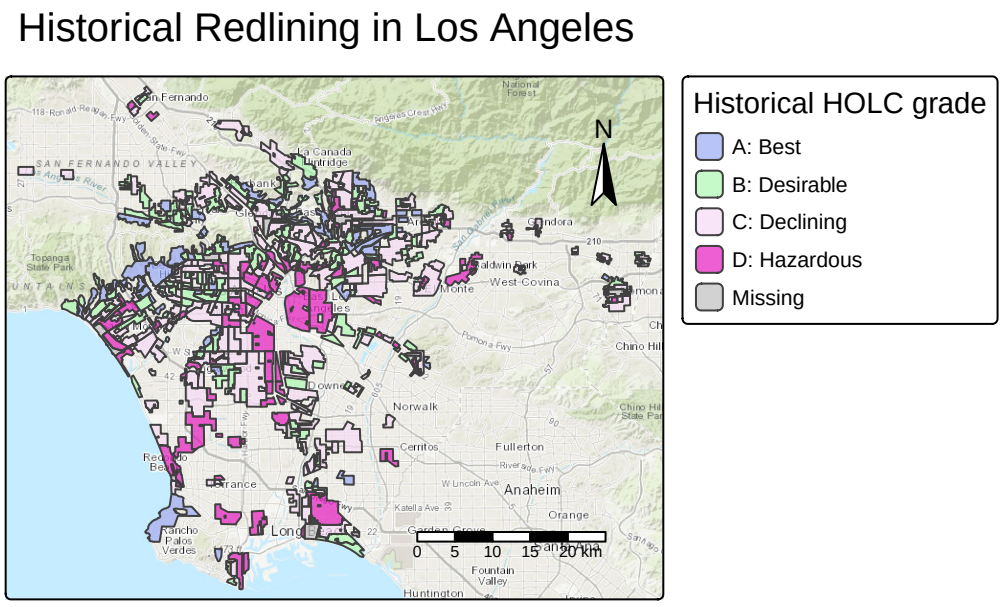
# HOLC grade colors
holc_colors <- c(
  "A" = "#99a9f6",
  "B" = "#adf9b2",
```

```
  "C" = "#f6dcf5",
  "D" = "#e619c0")

plot_colors <- c(
  holc_colors,
  "Not graded" = "#999999")

holc_map <- tm_basemap(server = "Esri.WorldTopoMap") +
  tm_shape(holc, is.main = TRUE) +
  tm_polygons(
    fill = "grade",
    fill_alpha = 0.7,
    fill.scale = tm_scale_categorical(
      values = holc_colors,
      labels = c("A: Best",
        "B: Desirable",
        "C: Declining",
        "D: Hazardous"
      )),
    fill.legend = tm_legend( title = "Historical HOLC grade",
    ))  +
  tm_compass(position = c("right", "top")) +
  tm_scalebar(position = c("right", "bottom")) +
  tm_title("Historical Redlining in Los Angeles")

holc_map # Display map
```



This map is showing us the neighborhoos and their corresponding HOLC grade, ranging from Best(A) to Hazardous (D)

```
# Filter ejscreen to LA County
ejscreen_filtered <- ejscreen |>
  filter(
    state_name == "California",
    cnty_name == "Los Angeles County"
  ) |>
  mutate(bg_id = row_number())

# Transform data to the same CRS
ejscreen_filtered <- ejscreen_filtered |>
  st_transform(3310) |>
  st_make_valid()

holc_transformed <- holc |>
  select(grade) |>
  st_transform(st_crs(ejscreen_filtered)) |>
  st_make_valid()

# Join HOLC grades to block groups
ej_holc_join <- st_join(
```

```
  ejscreen_filtered,
  holc_transformed,
  largest = TRUE,
  left = TRUE
)
```

```
# drop geometry and label block groups without a HOLC grade
dropped_geom <- ej_holc_join |>
  st_drop_geometry() |>
  mutate(
    grade = replace_na(as.character(grade), "Not graded"),
    grade = factor(grade, levels = c("A", "B", "C", "D", "Not graded")
    ))

# Calculate block counts by grade
grade_summ <- dropped_geom |>
  group_by(grade, .drop = FALSE) |>
  summarise(total_block_groups = n(), .groups = "drop") |>
  mutate(percentage = total_block_groups /sum(total_block_groups) * 100)

# Display table
kable(grade_summ, col.names = c(
    "HOLC Grade",
    "# of Block Groups",
    "% of Block Groups"),
  digits = 2,
  caption = "LA County census blocks by HOLC grade."
)
```

Table 1: LA County census blocks by HOLC grade.

HOLC Grade	# of Block Groups	% of Block Groups
A	261	3.96
B	664	10.07
C	2029	30.78
D	938	14.23
Not graded	2699	40.95

```
# Mean low income population
low_income <- dropped_geom |>
  group_by(grade) |>
```

```
  summarise(mean_low_income = mean(lowincpct, na.rm = TRUE) * 100,
  .groups = "drop")

# Mean PM2.5
pm25 <- dropped_geom |>
  group_by(grade) |>
  summarise(mean_pm25 = mean(p_pm25, na.rm = TRUE),
  .groups = "drop")

# Mean low life expectancy
low_life_exp <- dropped_geom |>
  group_by(grade) |>
  summarise(mean_lle = mean(p_lifeexppct, na.rm = TRUE),
  .groups = "drop")
```

::: {.cell}

```
ggplot(
low_income, aes(x = grade, y = mean_low_income, fill = grade)) +
geom_col(color = "black", width = 0.6) +
scale_fill_manual(values = plot_colors) +
labs(
  title = "Low Income Population by HOLC Grade",
  x = "HOLC Grade",
  y = "Mean Low Income Population Percent") +
theme_minimal() +
theme(
  legend.position = "none",
  panel.grid.major.x = element_blank()
)
```

:::

::: {.cell}

```
ggplot(pm25,
aes(x = grade, y = mean_pm25, fill = grade)) +
geom_col(color = "black", width = 0.6) +
scale_fill_manual(values = plot_colors) +
scale_y_continuous(limits = c(0, 100)) +
labs(
  title = "PM2.5 Percentile by HOLC Grade",
```

```
  x = "Assigned HOLC Grade",
  y = "Mean PM2.5 Percentile"
) +
theme_minimal() +
theme(
  legend.position = "none",
  panel.grid.major.x = element_blank())
```

:::

::: {.cell}

```
ggplot(low_life_exp,
aes(x = grade, y = mean_lle, fill = grade)) +
geom_col(color = "black", width = 0.6) +
scale_fill_manual(values = plot_colors) +
scale_y_continuous(limits = c(0, 100)) +
labs(
  title = "Low Life Expectancy by HOLC Grade",
  x = "Assigned HOLC Grade",
  y = "Mean Low-Life-Expectancy Percentile") +
theme_minimal() +
theme(
  legend.position = "none",
  panel.grid.major.x = element_blank())
```

:::

- Looking at the different plots for low-income populations, low life expectancy, and particualte matter, they are all highest in HOLC grade D areas, which is classified as "Hazardous", and are lowest in HOLC grade A areas, which is considered to be the "Best" areas. This pattern can lead us to believe that there are historical redlining that still have environmental and socioeconomic impacts on those that continue to live in these areas.

Part 2

```
# Clean column names
crashes <- crashes |>
  clean_names()
```

```r
# look for missing coordinate pairs
coordinate_missing<- crashes |>
  summarise(
    total_observations = n(),
    missing_point_xy = sum(is.na(point_x) | is.na(point_y)),
    missing_long_lat = sum(is.na(longitude) | is.na(latitude))
  )

kable(coordinate_missing)
```

total_observations	missing_point_xy	missing_long_lat
251100	9484	97350

```r
# Filter pedestrian or bicycle crashes
bike_ped_crashes <- crashes |>
  mutate(collision_date = lubridate::ymd(as.character(collision_date)),
    number_killed = as.numeric(number_killed)) |>
  filter(
    collision_date >= as.Date("2020-01-01"),
    collision_date <= as.Date("2025-12-31"),
    pedestrian_accident == "Y" | bicycle_accident == "Y")

# Remove missing coordinates
sf_crashes <- bike_ped_crashes |>
  drop_na(point_x, point_y) |>
  st_as_sf(coords = c("point_x", "point_y"),
    crs = 4326) |>
  st_transform(st_crs(holc_transformed))
```

```r
# Join crash points to HOLC grades
crashes_holc <- st_join(
  sf_crashes,holc_transformed |> select(grade))

# keep crashes in the HOLC areas
crashes_graded <- crashes_holc |>
  st_drop_geometry() |>
  filter(!is.na(grade))
```

```r
# Count crashes in each HOLC grade
most_crashes <- crashes_graded |>
  group_by(grade, .drop = FALSE) |>
  summarise(n_points = n(),.groups = "drop")

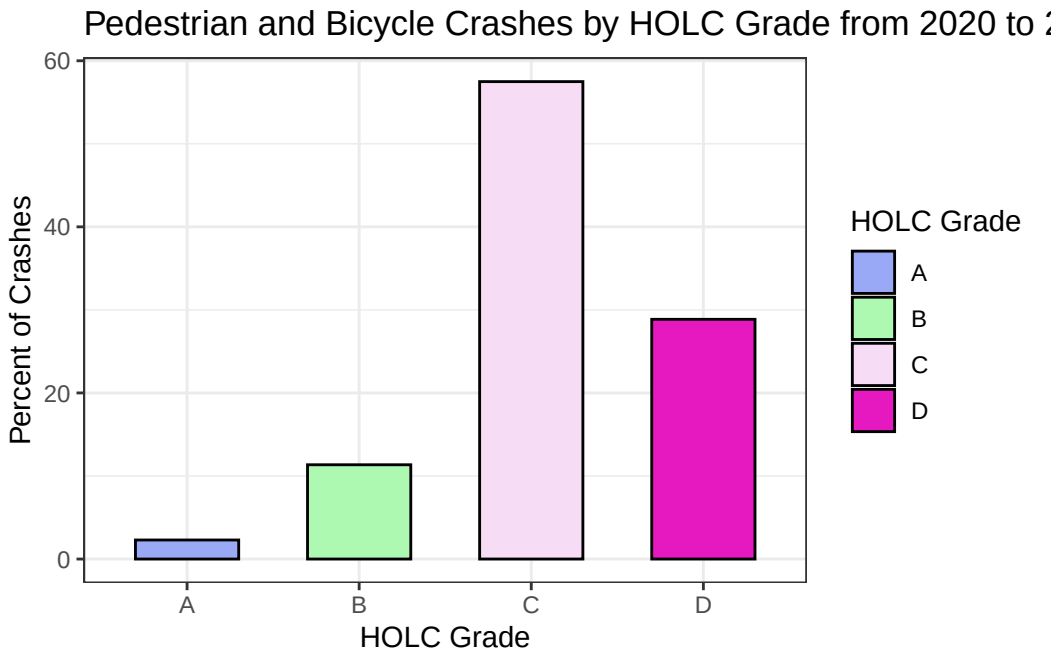
# Calculate the percentage in each grade
percent <- most_crashes |>
  mutate(percentage = n_points / sum(n_points) * 100)

kable(percent,
  digits = 1,
  caption = "Pedestrian and bicycle crashes by HOLC grade, 2020-2025.")
```

Table 2: Pedestrian and bicycle crashes by HOLC grade, 2020–2025.

grade	n_points	percentage
A	360	2.3
B	1782	11.4
C	9026	57.5
D	4532	28.9

```r
# Visualize crashes
ggplot(data = percent,
  mapping = aes(
    x = grade,
    y = percentage,
    fill = grade
  )) +
  geom_col(color = "black", width = 0.6) +
  scale_fill_manual(values = holc_colors) +
  theme_bw() +
  labs(
    title = "Pedestrian and Bicycle Crashes by HOLC Grade from 2020 to 2025",
    x = "HOLC Grade",
    y = "Percent of Crashes",
    fill = "HOLC Grade"
  )
```


Pedestrian and Bicycle Crashes by HOLC Grade from 2020 to 2
Percent of Crashes
60
40
20
0
A
B
C
D
HOLC Grade
HOLC Grade
A
B
C
D